VMC Vidyamandir Classes SINCE 1986
IIT JEE | MEDICAL | FOUNDATION

CODE A

JEE Main - 1 | JEE - 2022

Date: 27/07/2020 | **Maximum Marks: 300**

Timing: 05:00 PM - 08:00 PM | **Duration: 3.0 Hrs**

General Instructions

1. The test is of **3 hours** duration and the maximum marks is **300.**
2. The question paper consists of **3 Parts** (Part I: **Physics**, Part II: **Chemistry**, Part III: **Mathematics**). Each Part has **two** sections (Section 1 & Section 2).
3. **Section 1** contains **20 Multiple Choice Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.
4. **Section 2** contains **5 Numerical Value Type Questions**. The answer to each question is an **integer** ranging from 0 to 99 (both inclusive).
5. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
7. On completion of the test, the candidate must hand over the Answer Sheet to the **Invigilator** on duty in the Room/Hall. **However, the candidates are allowed to take away this Test Booklet with them**.
8. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Marking Scheme

1. **Section – 1:** +4 for correct answer, –1 (negative marking) for incorrect answer, 0 for all other cases.
2. **Section – 2:** +4 for correct answer, 0 for all other cases. There is no negative marking.

Name of the Candidate (In CAPITALS) : ________________________

Roll Number : ________________________

OMR Bar Code Number : ________________________

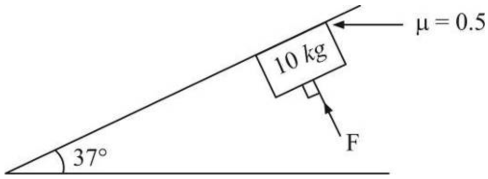
Candidate's Signature : ____________ **Invigilator's Signature** ____________

PART - I : PHYSICS — 100 MARKS

SECTION-1

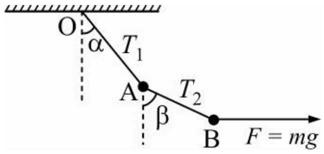
This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. In the figure shown, the minimum force F to be applied perpendicular to the incline so that the block does not slide is : {Take $g = 10$ m/s^2}



(A) 0 **(B)** 40 N
(C) 120 N **(D)** 200 N

2. Two particles A and B, each of mass m, are kept stationary by applying a horizontal force $F = mg$ on particle B as shown in figure, then :



(A) $2\tan\beta = \tan\alpha$ **(B)** $2T_1 = 5T_2$
(C) $T_1 = T_2$ **(D)** None of these

3. If $\vec{b} = 3\hat{i} + 4\hat{j}$ and $\vec{a} = \hat{i} - \hat{j}$, the vector having the same magnitude as that of $\vec{b}$ and parallel to $\vec{a}$ is :

(A) $\frac{5}{\sqrt{2}}(\hat{i} - \hat{j})$ **(B)** $\frac{5}{\sqrt{2}}(\hat{i} + \hat{j})$ **(C)** $5(\hat{i} - \hat{j})$ **(D)** $5(\hat{i} + \hat{j})$

4. A man running uniformly at 8 m/s is 16 m behind a bus when it starts accelerating at $2\ \text{ms}^{-2}$ from rest. Time taken by him to board the bus is :

(A) 2s **(B)** 3s **(C)** 4s **(D)** 5s

5. Consider the following statements :

I. A body has zero velocity and is still accelerating.
II. The velocity of an object reverses direction when acceleration is constant.
III. An object's speed goes on increasing even when magnitude of its acceleration decreases.

Which of the statements can be true?

(A) Only I **(B)** Only II **(C)** I, II and III **(D)** Only III

6. A body is projected with velocity u at an angle of projection θ with horizontal. The direction of velocity of the body makes angle $30°$ with the horizontal at t = 2s and then after 1s it reaches the maximum height. then: {Take $g = 10$ m/s^2 }

(A) $u = 30\sqrt{3} ms^{-1}$ **(B)** $\theta = 60°$ **(C)** $\theta = 30°$ **(D)** $u = 10\sqrt{3}\ ms^{-1}$

7. A particle moves along the parabolic path $y = ax^2$ in such a way that the x component of the velocity remains constant, say c. The acceleration of the particle is

(A) $ac\hat{k}$ **(B)** $2ac^2\hat{j}$ **(C)** $2ac^3\hat{j}$ **(D)** $a^2c\hat{j}$

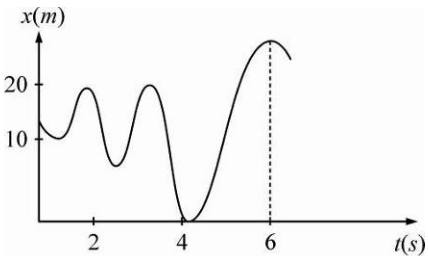
8. An object may have :

(A) Varying speed without having varying velocity
(B) Varying velocity without having varying speed
(C) Non-zero acceleration without having varying velocity
(D) Zero acceleration without having varying speed in a circular motion

9. A parachutist drops first freely from an aeroplane for 10*s* and then parachute opens out. Now he descends with a net retardation of 2.5 m/s^2. If he bails out of the plane at a height of 2495 *m* and $g = 10$ m/s^2, his velocity on reaching the ground will be :

(A) 5 *m/s* **(B)** 10 *m/s* **(C)** 15 *m/s* **(D)** 20 *m/s*

10. The graph shows the position of a particle moving on the *x*-axis as a function of time.



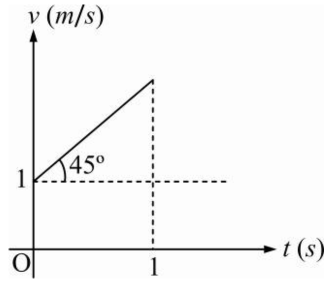
(A) The particle has come to rest 6 times

(B) The maximum speed is at $t = 6$ *s*

(C) The velocity remains positive for $t = 0$ to $t = 6$ *s*

(D) The average velocity of the total period shown is negative

11. From the *v*-*t* graph, the:



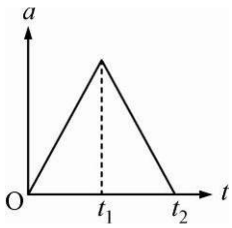
(A) Speed at $t = 1$ s is 1.2 *m/s*

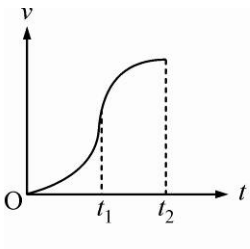
(B) Acceleration is 2 m/s^2

(C) Average speed during 1st second is 1.5 *m/s*

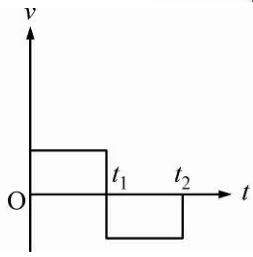
(D) Speed of the particle can be zero

12. The acceleration versus time graph of a particle starting from rest is shown in figure. The respective *v*-*t* graph of the particle is :

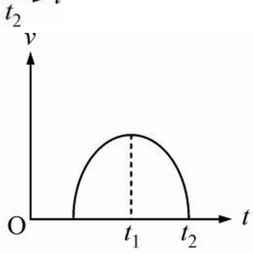




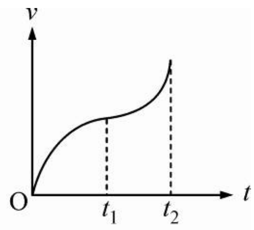
(A)



(B)

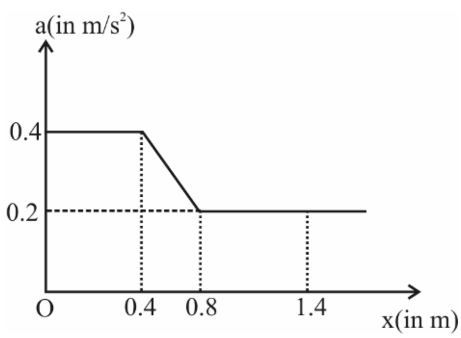


(C)



(D)

13. The acceleration of a particle which moves along the positive *x*-axis varies with its position as shown. If the velocity of the particle is 0.8 *m/s* at $x = 0$, the velocity of the particle at $x = 1.4$ m is (in *m/s*):



(A) 1.6

(B) 1.2

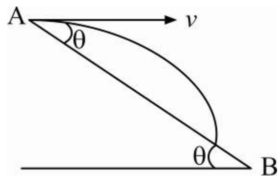
(C) 1.4

(D) None of these

14. The velocity of a body moving in a straight line is given by $v = (3x^2 + x)\ m/s$ (x is in meter). Find acceleration at $x = 2m$.

(A) $182\,m/s^2$ **(B)** $172\,m/s^2$ **(C)** $192\,m/s^2$ **(D)** $162\,m/s^2$

15. A particle is projected horizontally with a speed $v = 5\,ms^{-1}$ from the top of a plane inclined at an angle $\theta = 37°$ to the horizontal as shown in figure :



Find the time taken by the particle to hit the plane. {Take $g = 10\text{m/s}^2$ }

(A) 3/4 s **(B)** 3s

(C) 4s **(D)** 4/3s

16. A particle is moving with velocity $\vec{v} = k\left(y\hat{i} + x\hat{j}\right)$, where k is constant. The general equation for its path is {x and y are coordinates of particles at any instant}

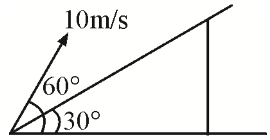
(A) $y = x^2 +$ constant **(B)** $y^2 = x +$ constant

(C) $xy =$ constant **(D)** $y^2 = x^2 +$ constant

17. A particle starts from rest at t = 0 and starts moving on a circular path of radius $\frac{4}{\pi}m$ with tangential acceleration $6t\ m/s^2$ where t is in seconds. Calculate acceleration of the particle when it completes one revolution in (m/s^2)

(A) $12\sqrt{16\pi^2+1}$ **(B)** $12\sqrt{1+9\pi^2}$ **(C)** $12\sqrt{16+\pi^2}$ **(D)** $12\sqrt{9+16\pi^2}$

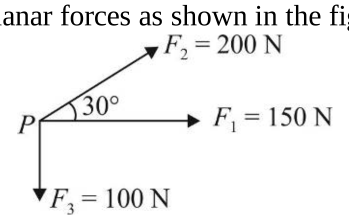
18. A particle is thrown at time $t = 0$ with a velocity of $10\,m/s$ at an angle of 60° with the horizontal from a point on an inclined plane making an angle 30° with horizontal. The time when the velocity of the projectile becomes parallel to the incline is : [Take g = 10 m/s^2]



(A) $\frac{2}{\sqrt{3}}sec$ **(B)** $\frac{1}{\sqrt{3}}sec$

(C) $\sqrt{3}\,sec$ **(D)** $\frac{1}{2\sqrt{3}}sec$

19. A particle P is acted by three coplanar forces as shown in the figure.



Find the force needed to prevent the particle P from moving. (take, $\sqrt{3} = 1.7$)

(A) 320 N in the direction of F_1 **(B)** 200 N in opposite direction of F_2

(C) 320 N in opposite direction of F_1 **(D)** 320 N at an angle 53° with direction of F_3

20. Find unit vector perpendicular to the plane of $a = 2\hat{i} - 2\hat{j} + \hat{k}$ and $b = \hat{i} + 2\hat{j} + 2\hat{k}$.

(A) $\frac{2}{3}\hat{i} + \frac{1}{3}\hat{j} + \frac{2}{3}\hat{k}$ **(B)** $-\frac{2}{3}\hat{i} - \frac{1}{3}\hat{j} - \frac{2}{3}\hat{k}$

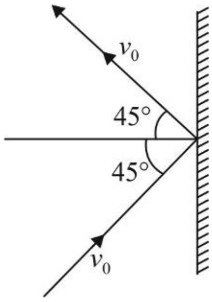
(C) $-\frac{2}{3}\hat{i} - \frac{1}{3}\hat{j} + \frac{2}{3}\hat{k}$ **(D)** $\frac{2}{3}\hat{i} + \frac{2}{3}\hat{j} - \frac{1}{3}\hat{k}$

SPACE FOR ROUGH WORK

SECTION-2

This section contains Five (05) Numerical Value Type Questions. The answer to each question is an integer ranging from 0 to 99 (both inclusive).

1. A particle moving with speed $v_0 = 5\sqrt{2}\ \text{ms}^{-1}$ strikes a wall at an angle 45° as shown in figure. If magnitude of change in velocity is $5x$ m/s, find x.



2. The initial velocity of a body moving along a straight line is 7 m/s. It has a uniform acceleration of $4\ \text{m/s}^2$, directed along initial velocity. The distance covered by the body in the 5th second of its motion is $5x$ metres. Find x.

3. The velocity-time relation of a particle starting from rest is given by $v = kt$ where $k = 2\ m/s^2$. The distance travelled in first 3s is x metres. Find x.

4. A ball is thrown vertically upwards with a speed of 10 m/s from the top of a tower 200 m high and another is thrown vertically downwards with the same speed simultaneously. The time difference between them in reaching the ground (g = 10ms^2) is x seconds. Find x.

5. If $\vec{a} = 2\hat{i} - 3\hat{j} + \hat{k}$ and $\vec{b} = x\hat{i} + \hat{j} + \hat{k}$ are mutually perpendicular. Find the value of x.

SPACE FOR ROUGH WORK

PART - II : CHEMISTRY 100 MARKS

SECTION-1

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. Which set of orbitals is listed in the sequential order of filling in a many-electron atom?

(A) 3s, 4p, 3d **(B)** 3d, 4s, 4p **(C)** 3d, 4p, 5s **(D)** 4p, 4d, 5s

2. All of the following possess complete d shells EXCEPT [Given: Atomic No. Cu = 29, Zn = 30, Ga = 31, Ag = 47]

(A) Ag^{+} **(B)** Cu^{2+} **(C)** Ga^{3+} **(D)** Zn^{2+}

3. Which element has the lowest first ionization energy?

(A) B **(B)** C **(C)** Al **(D)** Si

4. Which of these elements has the greatest electronegativity ?

(A) Br **(B)** N **(C)** O **(D)** S

5. When the atoms Li, Be, B and Na are arranged in order of increasing atomic radius, what is the correct order ?

(A) B, Be, Li, Na **(B)** Li, Be, B, Na **(C)** Be, Li, B, Na **(D)** Be, B, Li, Na

6. Which set is expected to show the smallest difference in first-ionization energy?

(A) He, Ne, Ar **(B)** B, N, O

(C) Mg, Mg^{+}, Mg^{2+} **(D)** Fe, Co, Ni

7. A chloride salt of rhenium contains 63.6% Re by mass. What is its empirical formula? (Atomic mass of Re : 186 amu, Cl = 35.5 amu)

(A) $ReCl$ **(B)** $ReCl_2$ **(C)** $ReCl_3$ **(D)** $ReCl_5$

8. According to the equation

$$SnO_2 + 2H_2 \rightarrow Sn + 2H_2O$$

What volume of hydrogen, measured at 1 atm and 273 K, is required to react with 2.00 g of SnO_2? (Atomic mass of Sn : 119 amu, O = 16 amu)

(A) 0.00133 L **(B)** 0.00265 L **(C)** 0.297 L **(D)** 0.593 L

9. When a phosphorus atom is converted to a phosphide ion, what happens to the number of unpaired electrons and the total number of electrons around the phosphorus ?

	Unpaired electrons	Total electrons
(A)	increases	increases
(B)	decreases	increases
(C)	increases	remains the same
(D)	decreases	remains the same

10. Which list includes elements in order of increasing metallic character?

(A) Si, P, S **(B)** As, P, N **(C)** Al, Ge, Sb **(D)** Br, Se, As

11. How do the energy gaps between successive electron energy level in an atom vary from low to high n values ?

(A) All energy gaps are the same
(B) The energy gap decreases as n increases
(C) The energy gap increases as n increases
(D) The energy gap changes unpredictably as n increases

12. $\underset{\text{salicylic acid}}{C_7H_6O_3} + \underset{\text{acetic anhydride}}{C_4H_6O_3} \rightarrow \underset{\text{aspirin}}{C_9H_8O_4} + \underset{\text{acetic acid}}{C_2H_4O_2}$

What is the percent yield if 0.85 g of aspirin is formed in the reaction of 1.00 g of salicylic acid with excess acetic anhydride ?

Substance	**Molar Mass**
$C_7H_6O_3$	$138.12\ g\ mol^{-1}$
$C_4H_6O_3$	$102.09\ g\ mol^{-1}$
$C_9H_8O_4$	$180.15\ g\ mol^{-1}$
$C_2H_4O_2$	$60.05\ g\ mol^{-1}$

(A) 65% **(B)** 77% **(C)** 85% **(D)** 91%

13. Which property of an element is most dependent on the shielding effect?

(A) atomic number **(B)** atomic mass
(C) atomic radius **(D)** number of stable isotopes

14. Given this set of quantum numbers for a multi-electron atom: 2, 0, 0, +1/2 and 2, 0, 0, –1/2. What is the next higher allowed set of n and ℓ quantum numbers for this atom in its ground state?

(A) $n = 2, \ell = 0$ **(B)** $n = 2, \ell = 1$ **(C)** $n = 3, \ell = 0$ **(D)** $n = 3, \ell = 1$

15. When the elements carbon, nitrogen and oxygen are arranged in order of increasing ionization energies, what is the correct order?

(A) C, N, O **(B)** O, N, C **(C)** N, C, O **(D)** C, O, N

16. Which pair of elements have chemical properties that are the most similar?

(A) Be and B **(B)** Al and Ga **(C)** Co and Cu **(D)** F and I

17. Which has the highest ionization energy for the removal of the second electron?

(A) F **(B)** Ne **(C)** Na **(D)** Mg

18. In a hydrogen atom, which transition produces a photon with the highest energy?

(A) $n = 3 \rightarrow n = 1$ **(B)** $n = 5 \rightarrow n = 3$ **(C)** $n = 12 \rightarrow n = 10$ **(D)** $n = 22 \rightarrow n = 20$

19. The emission spectrum of hydrogen in the visible region consists of

(A) a continuous band of light
(B) a series of equally spaced lines
(C) a series of lines that are closer at low energies
(D) a series of lines that are closer at high energies

20. Which of the following atomic orbitals contains one nodal plane?

(A) 3s **(B)** $3p_y$ **(C)** $3d_z 2$ **(D)** $3d_{xy}$

SPACE FOR ROUGH WORK

SECTION-2

This section contains Five (05) Numerical Value Type Questions. The answer to each question is an integer ranging from 0 to 99 (both inclusive).

1. $3.0\ dm^3$ of sulfur dioxide is reacted with $2.0\ dm^3$ of oxygen according to the equation given below.

$$2SO_2(g) + O_2(g) \rightarrow 2SO_3(g)$$

What volume of sulfur trioxide (in dm^3) is formed? (Assume the reaction goes to completion and all gases are measured at the same temperature and pressure.)

2. 6.0 moles of $Fe_2O_3(s)$ reacts with 9.0 moles of carbon in a blast furnace according to the equation given below.

$$Fe_2O_3(s) + 3C(s) \rightarrow 2Fe(s) + 3CO(g)$$

What is the theoretical yield of Fe in moles?

3. 1.50 g sample of an ore containing silver was dissolved, and all of the Ag^+ was converted to 1.24 g of Ag_2S. What was the percentage of silver in the ore? (Atomic mass of Ag : 108 amu, S: 32 amu)

4. How many orbitals contain one or more electrons in an isolated ground state iron atom (Z = 26)?

5. What is the sum of degeneracy of second excited state of hydrogen atom and H^- ion?

SPACE FOR ROUGH WORK

PART - III : MATHEMATICS 100 MARKS

SECTION-1

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. If $\sin t + \cos t = \frac{1}{5}$ then $\tan\frac{t}{2}$ is equal to $(\pi < t < 2\pi)$

(A) -1 **(B)** $-\frac{1}{3}$ **(C)** 2 **(D)** $-\frac{1}{6}$

2. If an angle A of a ΔABC satisfies $5\cos A + 3 = 0$, then the roots of the quadratic equation, $9x^2 + 27x + 20 = 0$ are :

(A) $\sin A, \sec A$ **(B)** $\sec A, \tan A$ **(C)** $\tan A, \cos A$ **(D)** $\sec A, \cot A$

3. If α, β, γ are the roots of $2x^3 - x + 1 = 0$ then the value of $\alpha^3 + \beta^3 + \gamma^3$ is :

(A) $-\frac{1}{2}$ **(B)** $\frac{1}{2}$ **(C)** $-\frac{3}{2}$ **(D)** $-\frac{5}{4}$

4. The real number x and y satisfy the equation $xy = \sin(2t)$ and $\frac{x}{y} = \tan(t)$ where $0 < t < \frac{\pi}{2}$. The value of $x^2 + y^2$, is:

(A) $\sqrt{2}$ **(B)** 1 **(C)** 2 **(D)** 4

5. $\frac{\sin\theta + \sin 2\theta}{1 + \cos\theta + \cos 2\theta}$ is equal to:

(A) $\sin\theta$ **(B)** $\cos\theta$ **(C)** $\tan\theta$ **(D)** $\cot\theta$

6. If $\cos A = \frac{3}{4}$, then $32\sin\left(\frac{A}{2}\right)\sin\left(\frac{5A}{2}\right) =$

(A) 7 **(B)** 8 **(C)** 11 **(D)** None of these

7. If α and β are roots of the equation $x^2 + 6x + \lambda = 0$ and $3\alpha + 2\beta = -20$, then $\alpha =$

(A) -8 **(B)** -16 **(C)** 16 **(D)** 8

8. If the roots of the equation $x^2 + px + q = 0$ are cubes of the roots of the equations $x^2 + mx + n = 0$, then :

(A) $p = m^3 + 3mn$ **(B)** $p = m^3 - 3mn$

(C) $p + q = m^3$ **(D)** $\frac{p}{q} = \left(\frac{m}{n}\right)^3$

9. The sum of the first and fifth terms of an AP is 26 and the product of the second and fourth is 160. Then the sum of the first six terms of the progression is :

(A) 59 or 69 **(B)** 69 or 87 **(C)** 87 or 109 **(D)** -69 or 87

10. Let $S = \{\theta \in [-2\pi, 2\pi] : 2\cos^2\theta + 3\sin\theta = 0\}$. Then the sum of the elements of S is :

(A) $\frac{13\pi}{6}$ **(B)** 2π **(C)** π **(D)** $\frac{5\pi}{3}$

11. a, b, c, d are in GP are in ascending order such that $a+d=112$ and $b+c=48.$ If the GP is continued with a as the first term, then the sum of the first six term is :

(A) 1156 **(B)** 1256 **(C)** 1356 **(D)** 1456

12. The sum $\sum_{k=1}^{20} k\frac{1}{2^k}$ is equal to :

(A) $2-\frac{11}{2^{19}}$ **(B)** $1-\frac{11}{2^{20}}$ **(C)** $2-\frac{21}{2^{20}}$ **(D)** $2-\frac{3}{2^{17}}$

13. Consider $3\sin\beta = \sin(2\alpha+\beta).$ Let K be the value of $\frac{\tan(\alpha+\beta)}{\tan\alpha}$ then the value of K is equal to :

(A) 0 **(B)** 2 **(C)** 4 **(D)** 8

14. The $1+\frac{1^3+2^3}{1+2}+\frac{1^3+2^3+3^3}{1+2+3}+....+\frac{1^3+2^3+3^3+....+15^3}{1+2+3.+....+15}-\frac{1}{2}(1+2+3+....+15)$ is :

(A) 620 **(B)** 1860 **(C)** 1240 **(D)** 660

15. If $-3<\frac{x^2+mx+1}{x^2+x+1}<3$ for all real $x,$ then :

(A) $m<-1$ **(B)** $-1<m<6$ **(C)** $-1<m<5$ **(D)** $m>6$

16. Three distinct numbers a, b, c from a GP in that order and the numbers $a+b, b+c, c+a$ form an AP in that order. Then the common ratio of the GP is :

(A) 1/2 **(B)** $-1/2$ **(C)** -2 **(D)** 2

17. Let $a_n, n \in N$ is an A.P. with common difference $`d'$ and all whose terms are non-zero. If n approaches infinity, then the sum $\frac{1}{a_1a_2}+\frac{1}{a_2a_3}+....+\frac{1}{a_na_{n+1}}$ will approach :

(A) $\frac{1}{a_1d}$ **(B)** $\frac{2}{a_1d}$ **(C)** $\frac{1}{2a_1d}$ **(D)** a_1d

18. The sum of all natural numbers $`n'$ such that $100<n<200$ and H.C. $F(91, n)>1$ is :

(A) 3221 **(B)** 3303 **(C)** 3203 **(D)** 3121

19. Let a, b and c in G.P. with common ratio $r,$ where $a \neq 0$ and $0<r\leq\frac{1}{2}.$ If $3a, 7b$ and $15c$ are the first three terms of an A.P., then the 4$^{\text{th}}$ term of this A.P. is :

(A) $\frac{2}{3}a$ **(B)** $\frac{7}{3}a$ **(C)** $5a$ **(D)** a

20. If $x^2+(a-b)x+1-a-b=0,$ where a and b are real numbers, has distinct real roots for all values of $b,$ then :

(A) $a<1$ **(B)** $a>1$ **(C)** $a<0$ **(D)** $0<a<1$

SPACE FOR ROUGH WORK

SECTION-2

This section contains Five (05) Numerical Value Type Questions. The answer to each question is an integer ranging from 0 to 99 (both inclusive).

1. Let x_1 and x_2 are the solution of the equation $\sec x = 1 + \cos x + \cos^2 x + \cos^3 x + ...\infty$ where $x_1, x_2 \in (0, 2\pi) - \{\pi\}$. The value of $|x_1 - x_2|$ is $\frac{a}{b}\pi$ (where a, b are coprime), then $a + b$ is _____ .

2. Maximum value of $\cos\left(x + \frac{\pi}{3}\right) + \cos x$ is a, then a^2 is____

3. Given that α, γ are roots of the equation $Ax^2 - 4x + 1 = 0$ and β, δ the roots or the equation of $Bx^2 - 6x + 1 = 0$, such that α, β, γ and δ are in H.P., then $A + B$ is :

4. Find the greatest integer value of c so that both the roots of the equation $(c-5)x^2 - 2cx + (c-4) = 0$ are positive, one root is less than 2 and other root is lying between 2 and 3.

5. If $\tan A$ and $\tan B$ are the roots of the quadratic equation, $3x^2 - 10x - 25 = 0$, then the value of $|3\sin^2(A+B) - 10\sin(A+B)\cdot\cos(A+B) - 25\cos^2(A+B)|$ is ______.

SPACE FOR ROUGH WORK

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